

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
K. K. BIRLA GOA CAMPUS

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SUPPLEMENTARY ANSWER BOOK

ID No. _____ Course No. _____

Name _____ Course Title _____

Date: _____

Start writing from here:-

Tejas & Advay

① Run Benchmarks on Dash Cluster.

② ~~SSD~~ → HDD vs SSD (N/W Bottleneck)

?? → Fast vs slow HDD
→ Fast vs slow SSD

③ BeeGP's placement algorithm.

?? → Experiments

??

By maths.

↳ Parameter Identification

↳ Values of the parameters

↳ Figured out N/W Bottleneck + local SSD + HDD results.

Motivation

① Does pooling even matter for low n/w condns.

② Does the state (Fast vs slow) matter when files are placed?

③ How does diff. placement algorithms function & what are the results??

To - DO:

① Intelligent Placement Algorithm that will understand the state of disks in real-time??

[Literature]
↓
[Literature]

Manoj

① Getting Application Patterns from Darshan Logs.

- ↳ Download all the ALCF logs.
- ↳ Parse those logs.
- ↳ Look at each log.

↓
classify each file in the log to different categories.

↓
small ~~read~~ vs med
vs
large
read vs write
contiguous vs random

read only
~~small~~
contiguous.

read-only
large read

Goal.

Class 1

Class 2

(user, process, app) App 1 →

2

2

rank	File 1	reads	lower
1	read 1K	→ 500	
	1M	→ 500	
	100	→ 1000	

File 2

App

File 1

MPI-IO APP

app = (nprocs = 10)

4 files.

File 1 (1.txt)
(Accessed by all 10 processes)
↓
Simple Shared File.
(SSF)

File 2 (2.txt)
↓
Accessed by process number 0
↓
File per process.
(FPP)

File 3 (3.txt)
↓
Accessed by process numbers
↓
File per process.
(CFPP)

File 4 (4.txt)
↓
Accessed by process numbers 0, 4, 6, 9
↓
Part-Shared File

Rank →

POSIX counters.
(84)

MPI-IO counters.
(50)

<u>POSIX</u>		84	50
rank	FN		
-1	1.txt	84	50
	read = 1000 1K = 1000		
0	2.txt	84	50
5	3.txt	84	50
0	4.txt	84	50
4	4.txt	84	50
6...8...9			

Week-wise Distribution

slides
+ github for code

May 4 - 8

- ① Download Darshan logs. & keep it in NAS.
- ② Make classes / categories of files from literature

May 11 - 15

APP1.csv

FI	-	-	-

- ① Take 1 darshan log.
- ② Script to parse the log. → csv.
- ③ Extract features & values, & keep it in the csv.

May 18 - 22

APP1.csv

CSV	class1	class2	nprocs	appid	usedid
APP1					
APP2					

- ① Take this csv. → classify each file in that csv to category
- ② Extend this for all darshan logs in 1 day.

May 25 - 29

- ① Extend this to 1 month of darshan logs.
- ② Start Analyzing 1 day . csv } script
(same APP. ⇒ behavior ??
same APP ⇒ diff. users.
same APP ⇒ diff. nprocs.)

June 1 - 5

- ① Extend analysis to 1 month.
- ② The most common patterns.
(which class is most frequent)

bar graph.
Heat map.

June 8 - 12

Buffer week.

June 15 - 19

① - Extend to 1 year.

② Do the analysis of

||

June 22 - 26

Continuing the analysis of
1 year duration logs.June 29 - Jul. 3

① Extend to 2nd year.

Jul 6 - 10

① Extend to 3rd year.

Jul 13 - 17. ① Grouping multi year.

Jul 20 - 24. ① Start writing a paper.

Jul 27 - 31

6-7 weeks.

End of Aug. ⇒ Paper submission

Qualifier : End of sep. / Early Oct.